

Multi Cloud **Generative** AI

DevSecOps

Job-Ready Curriculum Framework

Well-designed according to industry standards.

Curriculum

A seasoned **DevSecOps** professional with extensive hands-on expertise in real-world **cloud solutions**.



17+ Years of IT experience
MNC companies

Tech
Mahindra

9+ Years of in experience
Cloud and DevOps Platform

Capgemini 

6+ Years of Experience
Training & Mentoring

CSC

IBM

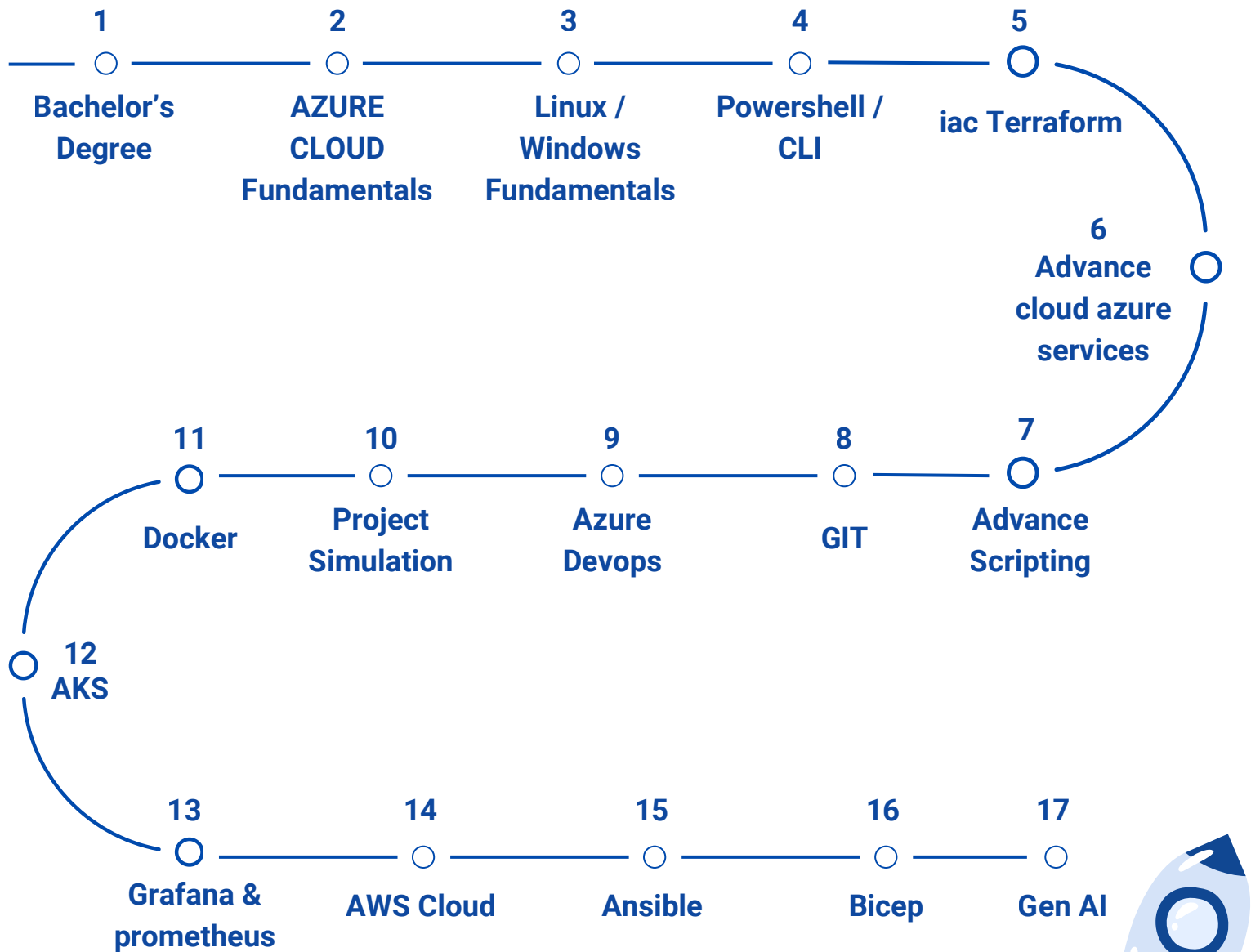
virtusa
Accelerating Business Outcomes



PRAVEEN G

Solution Architect, Technical Manager,
MCT Certified Trainer

ROAD MAP





అర్జునుడు & శ్రీకృష్ణ ం డు – విజయం వైపు మార్గదర్శనం

అర్జునుడు: "కృష్ణ ం , ఈ శ్రరరంచం వేగంగా మారిపోతూ ఉంది. నాలో సందేహం ఉంది, మండుకు ఎలా సాగాలి?"

శ్రీకృష్ణ ం డు: "అర్జునా, భయం నీ శత్రువు! నియమాలు తీసుకో, సాధన చెయ్, శ్రరయోగాలు చెయ్. విజయం నీ వంటే ఉంటుంది!"

అర్జునుడు: "కృష్ణ ం , కానీ ఎటవంటి ఫలితం వసుతందో తెలియదు! ఏదైనా తప్పు అయితే?"

శ్రీకృష్ణ ం డు: "అర్జునా, విను –

'కర్తవ్యే వాధికార్యత మా ఫలేష్ట కదాచన!'

నీ కర్తవ్యోనిర్వ రితంచు, కానీ ఫలంపై ఆశ పెటుకోకు. విజయం, రరాజయం అనేవి రండా ఒకే నాణ్యనికి రండు శ్రరకక లు. నిజమైన శక్తి త శ్రశమ, రటుదల, నిర్ంతర్ అభ్యే సంలో ఉంది. అనుభవించు, నూతనమై ఆలోచించు, భయానో విడిచపెటు!"

MODULE 1

CLOUD & AZURE FUNDAMENTALS

1.1 ORIENTATION SESSION:

KEY TOPICS:

- Welcome & Introduction
- Overview of Cloud Computing and DevOps
- Importance of Cloud and DevOps in modern businesses
- Benefits of cloud computing
- Why Organizations Migrating into Cloud
- Program expectations and outcomes
- Set Career Path on Cloud and DevOps

1.2 INTRODUCTION TO AI IN DEVOPS

KEY TOPICS:

- What is Generative AI?
- Role of AI in Software Development & IT Operations
- AI-powered DevOps lifecycle (Plan → Code → Build → Test → Release → Monitor)

1.3 AIOPS (AI FOR IT OPERATIONS)

KEY TOPICS:

- What is AIOps?
- How AIOps reduces IT Operations workload (log analysis, anomaly detection, auto-healing)
- Cloud-native AIOps platforms:

MODULE 1

CLOUD & AZURE FUNDAMENTALS

1.4 AI FOR INFRASTRUCTURE AS CODE (IAC)

KEY TOPICS:

- Using AI to generate Terraform/Ansible code
- How to validate and optimize IaC with AI tools
- Automating documentation of Terraform modules with AI

1.5 AI FOR KUBERNETES TROUBLESHOOTING

KEY TOPICS:

- Why K8s troubleshooting is hard (pods failing, networking issues, scaling problems)

HOW AI TOOLS HELP:

- Kubectl AI plugins
- K8sGPT – AI-powered troubleshooting for Kubernetes clusters
- Azure OpenAI + K8s logs for RCA (Root Cause Analysis)

1.6 GITHUB COPILOT & AI PAIR PROGRAMMING

KEY TOPICS:

- What is GitHub Copilot?
- How to set up Copilot for Powershell, Bash, Terraform, Kubernetes, and YAML files
- Prompt engineering for DevOps: writing better prompts for IaC & CI/CD

HOW AI TOOLS HELP:

- Generate Terraform code for an Azure Storage Account
- Write Kubernetes YAML manifests with Copilot
- Debug CI/CD pipeline YAML using Copilot

MODULE 1

CLOUD & AZURE FUNDAMENTALS

1.7 CLOUD COMPUTING FUNDAMENTALS:

KEY TOPICS:

- What is Cloud Computing?
- Cloud Service Models (IaaS, PaaS, SaaS)
- Cloud Deployment Models (Public, Private, Hybrid)
- Popular Cloud Providers (AWS, Azure, Google Cloud)
- Use cases and examples

1.8 AZURE OVERVIEW AND CORE SERVICES

KEY TOPICS:

- Azure Account creation
- AzTenant
- AzSubscriptions
- Azregions & Locations
- AzResource Groups
- AzAccount creation
- Architecture and Hierarchy
- Understanding IAAS, PAAS and SAAS

MODULE 2

AZURE NETWORKING

2.1 NETWORKING BASICS (IPADDRESS, SUBNETTING, VLAN, NAT...)

KEY TOPICS:

- IPAddress, Public and Private
- IP Address Classes and Subnetting
- DHCP, DNS
- LAN, WAN and VLAN
- NAT and PAT

2.2 IMPLEMENTING AND MANAGING AZURE NETWORKING

KEY TOPICS:

- Overview of Azure networking
- Implementing and managing Azure virtual networks
- Configuring Azure virtual networks
- Configuring Azure virtual network connectivity
- Configuring the Subnets
- Configuring Virtual Network Region Peering
- Configuring Virtual Network Global Peering
- Understanding Azure to On-Prem Connectivity
- Deploying Azure Virtual Network Gateway.
- Understanding User Defined Routes(UDR).
- Configuring Azure Virtual Network Gateway with AWS over IPsec VPN.
- Implementing Azure Service End Points.
- Understanding Hub and Spoke Architecture

2.3 UNDERSTANDING & CONFIGURING NETWORK SECURITY GROUPS (NSG)

KEY TOPICS:

- Understanding Azure Network Security Groups (NSG)
- Creating and Updating Inbound & Outbound Security Rules.
- Testing Network Security Groups and learning rule priority.
- Creating NSG Rules with Service tags.
- Understanding Application Security Groups (ASG).
- Creating NSG with Application Security Groups.

MODULE 3

AZURE POWERSHELL/BASH SCRIPTING

3.1 UNDERSTANDING AND START WRITING SCRIPT

KEY TOPICS:

- Cloud Shell
- Install AzModules on Laptop
- How to Write the Powershell or Bash scripts
- Execute and Deploy Bash script for deployment
- Prepare Azure CLI scripts to deploy LAB environment

3.2 IMPLEMENTING & CONFIGURING AZURE VIRTUAL MACHINES

KEY TOPICS:

- Overview Azure virtual machines
- Deploy virtual machines in Azure portal
- Deploy virtual machines using Azure CLI
- Managing Azure Virtual machine storage
- Understanding Availability sets, Fault Domain & Update Domain.
- Creating & place virtual machine in Availability Sets.
- Images, SizeDisk
- Troubleshoot VM Connectivity Issues
- Re-size VM
- Add and extent the Disk
- Hosting Web Application

3.3 UNDERSTANDING IDENTITY & GOVERNANCE

KEY TOPICS:

- Azure AD, Users, Groups, and Roles
- RBAC vs Azure Policies
- Management Groups, Subscriptions, Resource Groups
- Azure AD Connect (Basics)

WORK SHOP - I

🕒 Duration : 3hrs

GOAL

Provide a hands-on experience where students deploy and manage cloud infrastructure using Azure Portal and Scripting

ACTIVITIES :

- Prepare a Infrastructure as per the Architecture and requirement using Power CLI and Azure Portal
- Interactive Session with Students
- Conducting Soft Skills Programs
- Preparing the Interview questions



MODULE 4

TERRAFORM WITH AZURE

4.1 INTRODUCTION TO INFRASTRUCTURE AS CODE (IAC) AND TERRAFORM

KEY TOPICS:

- Introduction to Terraform
- Why Terraform in Cloud and DevOps
- Terraform Key Concepts and Fundamentals
- Terraform Programming Structure
- Terraform Architecture & Workflow
- Installing Terraform CLI
- Terraform with Visual Studio Code Setup

4.2 TERRAFORM BASICS

KEY TOPICS:

- Terraform Providers & Azure Provider Configuration
- Understanding HCL
- Writing the first Terraform configuration
- terraform init, plan, apply, and destroy

4.3 WORKING ON HIGH LEVEL BLOCKS OF TERRAFORM

KEY TOPICS:

- Working with Provider and Resource Blocks
- Variables and Outputs Blocks
- Data Sources Block
- Local Blocks
- Deploying Azure Resources (RG,vNet,NSGs,VM's Storage Accounts)
- What is Terraform State?
- Remote State Storage (Azure Storage Account as Backend)
- State Locking and Best Practices

WORK SHOP - 2

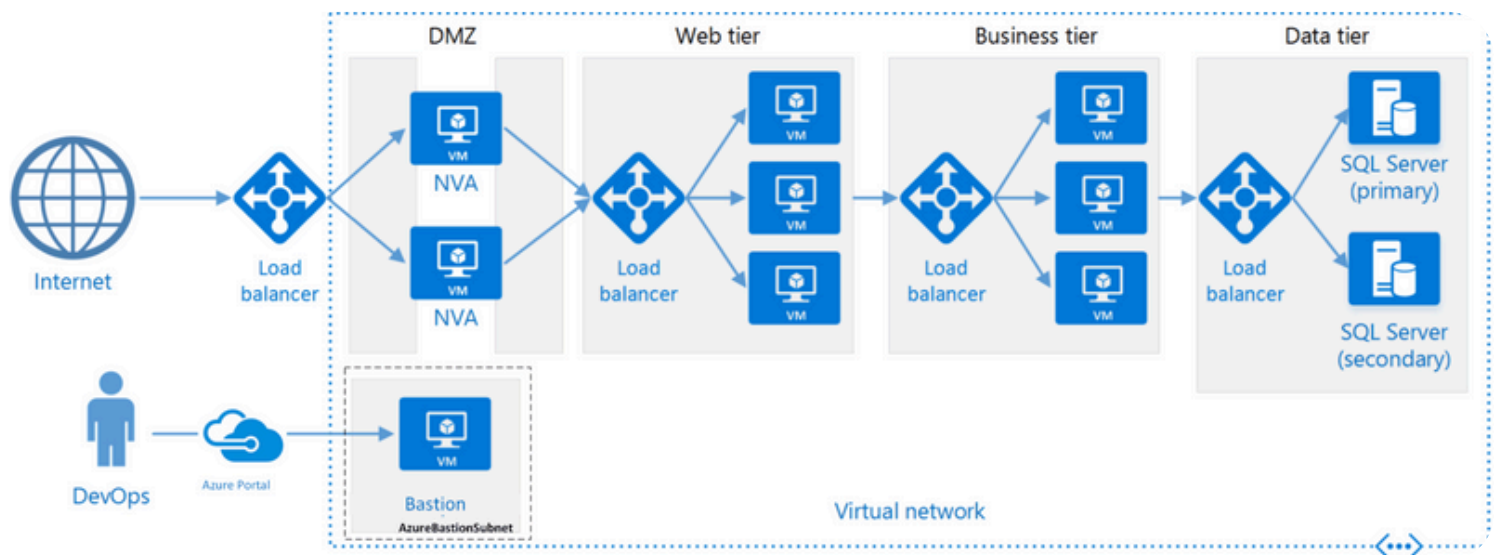
🕒 Duration : 3hrs

GOAL

Provide a hands-on experience where students deploy and manage cloud infrastructure using Terraform.

ACTIVITIES :

- Hands-on Lab with Terraform
- Install Terraform and set up a cloud provider (e.g., AWS, Azure)
- Based on the Architecture design, Students needs tto provision the Infrastructure.
- Execute Terraform commands: Terraform init, validate, plan, Apply and Destroy
- Manage the Terraform state and understand drift detection



MODULE 5

HIGH AVAILABILITY & AZURE LOAD BALANCERS

5.1 INTRODUCTION TO LOAD BALANCING IN AZURE

KEY TOPICS:

- Why Load Balancing?
- Different Load Balancing Solutions in Azure
- Overview on Load Balancing
- Type of Load Balancers in Azure
- Deploying Basic Load Balancer with Availability Sets.
- Checking the Load Balancing scenarios.
- Difference Between Azure Availability Sets and Availability Zone

5.2 DESIGNING & IMPLEMENTING AZURE LOAD BALANCING

KEY TOPICS:

- Basic vs. Standard Load Balancer
- Components: Frontend IP, Backend Pool, Health Probes, Rules
- Configuring a Public & Internal Load Balancer
- Cross-region Load Balancing
- Checking the Load Balancing scenarios.
- Configuring NAT using Basic Load Balancer

5.3 IMPLEMENT AND CONFIGURE AZURE DNS & AZURE STANDARD LOAD BALANCER

KEY TOPICS:

- Understanding Azure DNS.
- Buying a domain name for Godaddy.com
- Creating Azure DNS Zone and configure with Godaddy.
- Creating 3 Availability Sets and place single virtual machine in each AVSet.
- Create Azure Standard Load Balancer and create backend pool and assign all the 3 AVSets.
- Check the Load Balancing between the servers in different AV-Sets.
- Create Host Records for Server and check the name resolution.
- Create the CNAME record for the Load Balancer and check the website resolution

MODULE 5

HIGH AVAILABILITY & AZURE LOAD BALANCERS

5.4 APPLICATION GATEWAY (LAYER 7 LOAD BALANCER)

KEY TOPICS:

- Understanding Azure Application Gateway architecture.
- Understanding Path Based Routing in AGW.
- Creating multiple VMs and configuring application gateway.
- Configuring the SSL Off-loading on Application gateway.
- Check pathing based routing using AGW.
- Features: SSL Termination, URL-based Routing, WAF Integration
- Web Application Firewall (WAF) for Security

5.5 AZURE TRAFFIC MANAGER (DNS-BASED LOAD BALANCER)

KEY TOPICS:

- Routing Methods: Priority, Weighted, Performance, Geographic
- Global Load Balancing & Multi-Region Failover

5.6 AZURE FRONT DOOR (GLOBAL HTTP LOAD BALANCING)

KEY TOPICS:

- CDN & Performance Acceleration
- Global Traffic Distribution with Rules Engine
- Securing with WAF & Custom Domain Setup

WORK SHOP - 3

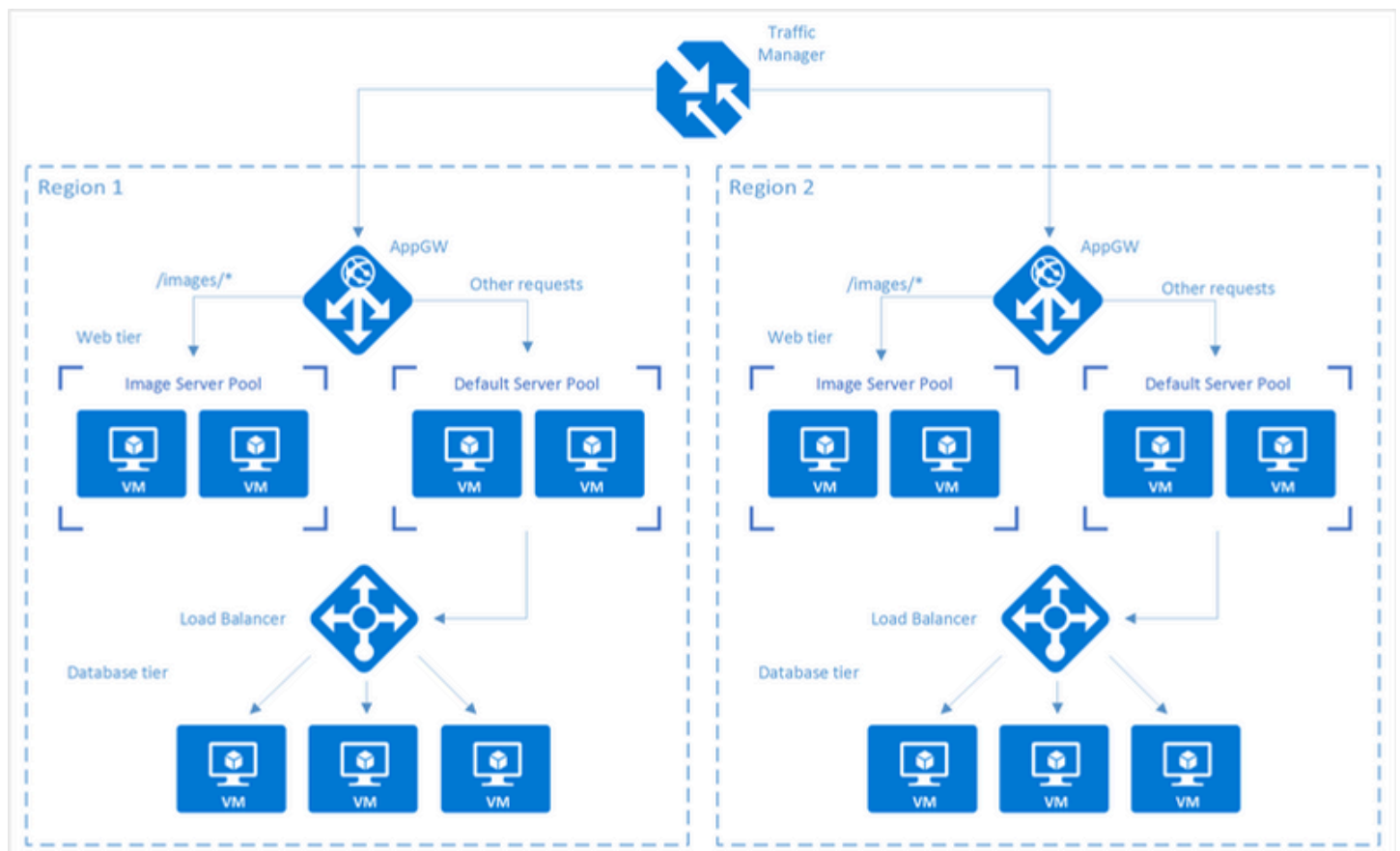
🕒 Duration : 3hrs

GOAL

End to End Infrastructure Deployment using Terraform and avail the concepts of Azure Load Balancers

ACTIVITIES :

- Hands-on Lab with Terraform
- Preparing the architecture Diagrams
- Prepare Module Level Code based on requirement/Architecture
- Practice the Load Balancer Topics
- Interview question Preparation
- Software skill Development



MODULE 6

PLATFORM AS A SERVICE(PAAS)

6.1 CONFIGURING AUTO SCALING WITH VIRTUAL MACHINE SCALE SETS(VMSS)

KEY TOPICS:

- Understanding Azure VMSS
- Creating a Azure VM Image for VMSS
- Deploy VMSS with the custom image
- Performing stress testing on the VMSS
- Check AutoScaling of VMs under stress testing

6.2 AZURE STORAGE ACCOUNT

KEY TOPICS:

- Azure Storage account overview
- Understand Blob Storage
- Understand File Shares
- Configuring Azure FileSync
- Data migration using Azure storage explorer
- Manage Azure Storage permissions
- Azure Static Website deployment
- Custom domains

6.3 AZURE BASTION(SECURE REMOTE ACCESS TO VIRTUAL MACHINES)

KEY TOPICS:

- Purpose & Benefits
- How Bastion Enhances Security by Eliminating Public Ips
- Bastion Standard vs. Basic Tiers

MODULE 6

PLATFORM AS A SERVICE(PAAS)

6.4 CONFIGURE & MANAGE AZURE BACKUP AND DISASTER RECOVERY SERVICES

KEY TOPICS:

- Backup and disaster recovery overview
- Backup VMware Servers to Azure
- Backup Azure VM & Azure SQL instances
- Azure replication and failover groups
- Azure disaster recovery vault configuration
- How to implement full BCDR strategy

6.5 PLANNING AND IMPLEMENTING AZURE SQL DATABASE

KEY TOPICS:

- Azure SQL database(PaaS) vs SQL database (IaaS)
- Structured vs Unstructured data
- DTUs - Azure SQL
- Azure SQL Global Replication and Failover Groups

6.6 AZURE KEY VAULT

KEY TOPICS:

- Introduction to Key vault
- Vault creation and management
- Adding secrets and Credentials
- Vault integration and Azure Pipelines

MODULE 6

PLATFORM AS A SERVICE(PAAS)

6.7 IMPLEMENTING AZURE ACTIVE DIRECTORY

KEY TOPICS:

- Azure Domain services
- Azure Active Directory overview
- Deploy and configure Active directory
- Create and manage users and groups
- Extending On-premise Active directory to Azure
- Configure Azure AD connect
- Role based access control (RBAC) overview
- Configure domain users and groups to access Azure portal

6.8 WEB APPS ON AZURE APP SERVICE

KEY TOPICS:

- App Service Plan (Basic, Standard, Premium)
- Sample Code Deployment
- Deployment Slots (Blue-Green Deployment)
- Custom Domains & SSL Certificates
- Auto-scaling Web Apps (Scale Up vs. Scale Out)
- Configuring Application Insights for Monitoring

6.7 AZURE FUNCTIONS APP AND LOGIC APP

KEY TOPICS:

- What is Azure Function App and Azure Logic App?
- Use Cases of Azure Function
- Use Cases of Logic App: Automation workflows, integration with APIs, email triggers
- Consumption Plan vs Premium vs Dedicated Plan
- Integration with Storage Account, Application Insights, and VNETs

MODULE 7

TERRAFORM ADVANCE TOPICS

7.1 TERRAFORM

KEY TOPICS:

- Terraform State Management
 - Remote State Storage
 - State Locking & Consistency
- Terraform Multiple Environments
- Terraform Modules & Reusability
 - Writing Reusable Terraform Modules
 - Structuring Terraform Code for Large-Scale Deployments
- Terraform Dynamic Blocks & Loops
 - Dynamic Blocks for Resource Reusability
 - For_each vs. Count in Resource Provisioning
 - Conditional Expressions & Null Resources

WORK SHOP - 4

 Duration : 3hrs

GOAL

End to End Infrastructure Deployment using Terraform Advance
Concepts & Pass Services

ACTIVITIES :

- Hands-on Lab with Terraform
- Preparing the architecture Diagrams
- Prepare Module level Code based on requirement/Architecture
- Interview question Preparation
- Software skill Development

MODULE 8

FUNDAMENTALS DEVOPS

8.1 INTRODUCTION OF DEVOPS & UNDERSTANDING OF DEVOPS METHODOLOGY

KEY TOPICS:

- History & Evolution of DevOps
- Differences Between Traditional IT, Agile, and DevOps
- DevOps Principles & Culture
- Benefits of DevOps for Organizations

8.2 DEVOPS LIFECYCLE & KEY PRACTICES

KEY TOPICS:

- Architecture of Continuous Integration (CI) & Continuous Deployment (CD)
- Understanding the Different Stages of CI and CD
- Different Automation tools which we can use in CI and CD

8.3 FUNDAMENTALS OF AZURE DEVOPS

KEY TOPICS:

- Understanding various Azure DevOps Services
- Setting up Azure DevOps free tier account
- Azure Project creation
- Understanding Azure DevOps Overview TAB
- Scenario-based interview question and solution
- Azure Services
 - Boards
 - Azure Repo
 - Pipelines
 - Test Plans
 - Artifacts

MODULE 8

FUNDAMENTALS DEVOPS

8.4 AZURE BOARDS & WORK ITEM MANAGEMENT

KEY TOPICS:

- Understanding Azure DevOps Boards
- Work item creation
- Backup and migration of work items
- Setting up iteration and path
- Customization Boards view

8.5 WORKING WITH GIT

KEY TOPICS:

- What is Git?
- Git vs GitHub vs GitLab vs Azure Repos
- Local Git Repositories vs Remote Repositories
- Basic Git Commands
 - git init, clone, status, add, commit, push, pull, fetch
 - git diff, log, reset, revert, rm, mv

8.6 REPOSITORY MANAGEMENT

KEY TOPICS:

- Introduction to Repository Management
- Centralized version control
- Distributed version control
- Understanding various stages of GIT
- Running through basic commands
- Understanding Advanced GIT Concepts
 - Cherry-picking
 - Rebase
 - Stash etc.

MODULE 8

FUNDAMENTALS DEVOPS

8.7 AZURE SERVICE CONNECTION:

KEY TOPICS:

- Introduction to Azure Service connections
- Understanding service principal concept
- Service connection creation

8.8 AZURE AGENT POOLS:

KEY TOPICS:

- Introduction to Agent pools
- Hosted vs Self-Hosted Pools
- Self-Hosted agent pool creation
- Integration of self-hosted agent pool with release pipelines

8.9 YAML FUNDAMENTALS:

KEY TOPICS:

- YAM Basic Syntax
- Data Types supported by YAML
- A Format and indentation of YAML

MODULE 8

FUNDAMENTALS DEVOPS

8.10 DEEP DIVE INTO AZURE DEVOPS PIPELINES

KEY TOPICS:

Azure Build Pipelines(CI)

- Build Pipelines
- Understanding continuous integration concepts
- Introduction to build process
- Project(1) – creation of ASP.NET Web application
- Classic build pipeline creation for ASP.Net application
- YAML build pipeline creation for ASP.NET application
- Different types of build triggers
- Pipelines custom variables
- Branch and Path filtration
- Understanding Pre-defined variables
- Project(2) – Introduction to Java Spring boot web application
- YAML build Pipeline creation for JAVA web application
- Project(3) – Introduction to SQL database automation
- Creation of SQL Database dacpac project
- YAML build Pipeline creation of SQL database project
- Deployment Groups
- Environments
- Pipeline Library
- Task Groups
- Scenario-based Interview Question with solution

MODULE 8

FUNDAMENTALS DEVOPS

8.11 RELEASE PIPELINES(CD):

KEY TOPICS:

- Understanding Continuous Deployment and Delivery concepts
- Introduction to the release process
- Classic release pipeline creation to deploy ASP.NET web Application
- Multi-Stage deployment Pipeline creation
- YAML release Pipeline creation to deploy JAVA Web Application
- Integration of JUNIT analysis with JAVA build pipelines
- Classic release pipelines creation to deploy SQL DACPAC packages
- Deployment Strategies in Azure DevOps
- Scenario-based Interview Question with Solution

8.12 MAVEN:

KEY TOPICS:

- Introduction to Maven build tools
- Maven benefits over others Grandle, Ant
- Maven commands
- Integration of Maven with JAVA spring boot build pipelines

8.13 AZURE ARTIFACTS:

KEY TOPICS:

- Introduction to Azure Artifacts
- Creation of Private feeds
- Integration of Private feeds in build pipelines
- Packaging and push the dependency in private feeds

MODULE 8

FUNDAMENTALS DEVOPS

8.14 AZURE SETTING:

KEY TOPICS:

- Introduction to Organization and Project settings
- Azure Active Directory Integration with DevOps
- Azure Permission
- User and Group management
- Retention and storage policies

8.15 BUILD PIPELINE SECURITY:

KEY TOPICS:

SonarQube (SAST):

- Introduction to SonarQube
- Setting up SonarQube in Docker container
- Downloading SonarQube plugins in Azure DevOps
- Creation of Service connection to connect Sonar Server
- Creation of New Project in SonarQube
- Integration of Sonar Analysis with Java Build pipelines
- Understanding publish gates and publish profiles
- Creation of custom publish gates
- Introduction custom publish gates in build pipelines
- Adding Sonar Break task
- Analyzing Sonar Build reports

MODULE 8

FUNDAMENTALS DEVOPS

8.16 TRIVY :

KEY TOPICS:

- Configuration of Trivy on ubuntu
- Scanning images using Trivy

8.17 CHECKOV FOR TERRAFORM

KEY TOPICS:

- Configuring Checkov for terraform
- Scanning IAC using Checkov

8.18 INFRASTRUCTURE AS CODE (IAC) WITH AZURE DEVOPS

KEY TOPICS:

- Terraform in Azure DevOps Pipelines
- Deploying Infrastructure using Azure DevOps & Terraform
- Implementing Security & Compliance in IaC

MODULE 9

ADVANCED PAAS AND CONTAINERIZATION

9.1 DOCKER ENGINE INSTALLATION

KEY TOPICS:

- Manual deployment of Docker CE on Ubuntu and CentOS
- Automate deployment with help of Vagrant Basic introduction to Docker Engine
- Different between Desktop and Server
- version check
- default locations
- commands execution

9.2 DOCKER ARCHITECTURE

KEY TOPICS:

- Docker daemon
- Docker client
- Docker registry
- Docker objects
- Images
- containers
- Networking
- Services

9.3 IMAGES AND CONTAINERMANAGEMENT

KEY TOPICS:

- docker image registries
- docker image pull/push
- container creation
- Sample application deployment like web server

MODULE 9

ADVANCED PAAS AND CONTAINERIZATION

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KEY TOPICS:

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- docker image pull/push
- container creation
- Sample application deployment like web server

MODULE 10

AWS CLOUD

10.1 FUNDAMENTALS AND NETWORKING IN AWS

KEY TOPICS:

- Introduction to AWS
- Networking Fundamentals
- VPC Overview
- VPC Components
- VPC Peering
- Creating and Configuring VPCs
- VPN Connections
- Transit Gateway
- Security Groups and Network ACLs
- Load Balancers
- AWS Route 53 Overview
- DNS Configuration
- Setting Up DNS
- AWS CloudWatch and Cloud Trail
- VPC Flow Logs

10.2 IAM [IDENTITY ACCESS MANAGEMENT]

KEY TOPICS:

- Introduction to AWS
- Introduction to IAM u IAM Components
- Creating IAM Users
- Password Policies
- Multi-Factor Authentication (MFA)
- Managing IAM Users
- Creating IAM Groups u IAM Roles
- Permissions Policies
- Group and Role Management
- IAM Policy Conditions
- Identity Federation
- Cross-Account Access

MODULE 10

AWS CLOUD

- Implementing Advanced IAM Scenarios
- IAM Best Practices u IAM Access Analyzer
- Introduction to AWS Organizations
- Service Control Policies (SCPs)
- Cross-Account IAM Roles

10.3 COMPUTE

KEY TOPICS:

- Introduction to EC2 instance
- Creating Launch templates
- Creating EC2 instance
- Saving sessions into putty
- Creating Images from EC2 instances

10.4 STORAGE

KEY TOPICS:

- Introduction to AWS Storage Services
- AWS S3 (Simple Storage Service)
- AWS EBS (Elastic Block Store)
- Amazon S3 Object Storage
- S3 Data Management
- S3 Security and Access Control
- Attaching and Detaching EBS Volumes
- EBS Snapshots
- Introduction to AWS Storage Gateway
- Data Lifecycle Policies
- S3 Glacier

MODULE 10

AWS CLOUD

10.5 DATABASES

KEY TOPICS:

- Introduction to AWS Database Services
- AWS RDS (Relational Database Service) Basics
- Relational Databases Overview
- Creating and Managing RDS Instances
- High Availability and Failover
- Data Backup and Disaster Recovery
- Introduction to Amazon Aurora
- Amazon Aurora vs. Traditional Databases
- Monitoring and Performance Optimization
- Cost Optimization

10.6 MANAGEMENT AND GOVERNANCE

KEY TOPICS:

- Introduction to AWS Management and Governance Services
- AWS Organizations
- AWS Control Tower
- AWS Cost Explorer
- AWS Budgets
- AWS Systems Manager
- AWS Config Rules
- AWS Organizations Policies

10.7 GOOGLE CLOUD

KEY TOPICS:

- What is Google Cloud Platform (GCP)?
- Key Differences Between GCP, AWS, and Azure
- GCP Global Infrastructure (Regions, Zones, Edge Locations)
- Understanding GCP Free Tier & Pricing Model
- Setting Up a GCP Account & Project
- Compute Services & Networking Services
- Storage Services

MODULE 10

AWS CLOUD

10.8 BICEP SCRIPTING

KEY TOPICS:

- What is Bicep & Why Use It?
- Bicep vs. ARM Templates vs. Terraform
- Setting Up the Bicep CLI & VS Code Extension
- Writing & Deploying Your First Bicep File
- Understanding Resource Declarations & Parameters
- Defining Variables & Parameters
- Using Modules for Reusability
- Looping & Conditional Deployments
- Working with Outputs & Expressions
- Best Practices for Writing Clean & Efficient Bicep Code

10.9 ANSIBLE

KEY TOPICS:

- What is Ansible & Why Use It?
- Agentless Architecture & How It Works
- Installing & Setting Up Ansible on Linux & Windows
- Ansible Inventory & Ad-Hoc Commands
- Running Your First Ansible Playbook
- Understanding YAML & Playbook Structure
- Defining Tasks, Modules, and Handlers
- Variables & Templates (Jinja2)
- Loops & Conditionals in Playbooks
- Debugging & Testing Ansible Playbooks

MODULE 10

AWS CLOUD

10.11 GRAFANA AND PROMETHEUS FUNDAMENTALS

KEY TOPICS:

- Why Monitoring is Important in Cloud and DevOps
- Traditional Monitoring vs Modern Cloud Monitoring
- Introduction to Metrics, Logs, and Tracing
- What is Prometheus?
- How Prometheus Works:
- What is Grafana?
- How Grafana Works:
- Install Prometheus and Node Exporter
- Configure Prometheus to scrape metrics
- Install Grafana
- Connect Grafana to Prometheus
- Create your First Dashboard (CPU Usage, Memory Usage)
- Set a Basic Alert on High CPU Usage

GUEST SPEAKER/INDUSTRY EXPERT TALK

 Duration : 3hrs

GOAL

- Deploying Applications to AKS using Azure DevOps Pipelines
- Implementing CI/CD for Web Apps using Azure DevOps Pipelines

ACTIVITIES :

- Hands-on Lab with AzureDevOps and AKS
- Sample Dotnet/Java application Project and Deploy using CI/CD
- Talk by a Cloud/DevOps professional
- Real-life examples and use cases in the industry
- Q&A session

CONCLUSION & FEEDBACK

This program has not only introduced you to Cloud and DevOps — it has **immersed you in real-world scenarios, enterprise-scale projects, and multi-cloud environments.**

- You've worked on Azure, AWS, and GCP just as cloud engineers do in real companies.
- You've automated infrastructure with Terraform, Ansible, and CI/CD pipelines like a seasoned DevOps engineer.
- You've deployed and managed containers and Kubernetes clusters in production-grade setups.
- You've explored how AI-powered tools (Copilot, AIOps, K8sGPT) transform modern IT operations.
- Your journey here is **not about basics**, it's about experiencing the role of a **3–5 year experienced Cloud & DevOps Engineer**, within a structured, guided learning path.

Personalized feedback from trainers ensures you know where you excel and what to strengthen further, giving you a **career roadmap to grow faster than your peers.**

PROGRAM OUTCOMES

Upon completion of this program, you will:

- Work at an industry-expert level across Azure, AWS, and GCP, not just with theory but with hands-on, real-time project experience.
- Be skilled in Terraform, Azure DevOps, Docker, Kubernetes, Ansible, Monitoring (Grafana/Prometheus), exactly what companies demand today.
- Harness AI for DevOps, use Copilot for IaC generation, AIOps for smarter operations, and AI-based troubleshooting for Kubernetes.
- Build end-to-end multi-cloud architectures and CI/CD workflows just like professionals working in large-scale enterprises.
- Graduate with a portfolio of 20+ projects that mirrors 3–5 years of industry expertise, making you job-ready from Day 1.
- Have a clear career path with certification guidance, placement or internship support, and the ability to confidently target roles like:
 - o Cloud Engineer
 - o DevOps Engineer
 - o Site Reliability Engineer (SRE)
 - o Cloud Automation Specialist

PROJECTS ARE BASED ON AWS/AZURE/DEVOPS

Note: Will regularly update the Projects based on the Industry ready requirement.

- Project-1 | 3-Tier Project on NSG
- Project-2 | 3 Tier project using Both Public and Private Load balancers with
- Project-3 | Creating Images using Packer both Windows and Linux
- Project-4 | Creating Completed Infrastructure (VM) using Terraform using Custom images
- Project-5 | Creating Landing Zone with Terraform
- Project-6 | Integrating both AWS and Azure cloud using VPN
- Project-7 | Project on Active Directory, users and servers will be part of Active Directory and all users authenticate to servers using Domain users
- Project-8 | Kubernetes Multi-Tier Capstone Mega Project
- Project-9 | Helm Chart Multi-Tier Mega Project
- Project-10 | Blue-Green Deployment
- Project-11 | 11 Microservices Project Deployment
- Project-12 | Dotnet Monitoring Project
- Project-13 | Multi-Tier NodeJS + MySQL Project
- Project-14 | Multi-Tier DotNET + MongoDB Project
- Project-15| Multi-Tier Python + Postgres Project
- Project-16 | Monitoring Projects with Kubernetes, Prometheus, Grafana
- Project-17 | Terraform & Ansible Corporate Project
- Project-18 | HashiCorp Vault Project
- Project-19 | App Deployment To Tomcat Server
- Project-20 | Azure DevOps Project | Multitier NodeJS+ DataBase Project Deploy To Webapp Using Classic Pipeline & Yaml Pipeline]
- Project-21 | Azure DevOps Project | Multitier Java Project Deploy To AKS
- Project-122 | Azure DevOps Project | Deploy 11-Microservice Based Application To Prod & Dev
- Project-23 | Python Monitoring Project
- Project-24 | DevSecOps Project- Complete End to End Mega project using Security tools in DevOps Pipeline

Thank You

For Your Attention



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